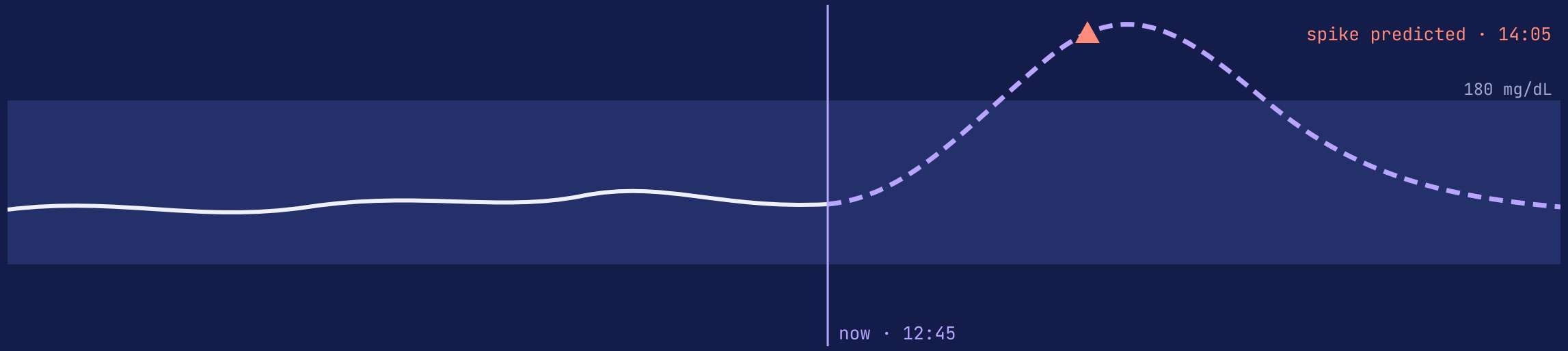
DIGITAL TWIN CHALLENGE 2026 · HAPPIEST HEALTH · SIMULATED DEMO

CHAYA छाया

The shadow that walks two hours ahead: a personal digital twin that sees a glucose spike **2 hours** before it happens, and puts the doctor and the patient on the same page.



THE PROBLEM

India's diabetes care runs on one number every three months.

101 M

Indians living with diabetes, plus 136 M with prediabetes

ICMR-INDIAB · Lancet D&E · 2023

62%

of calories in Indian diets come from carbohydrates

ICMR-INDIAB · Nature Medicine · 2025

~270

meals between two quarterly HbA1c tests, none of them seen by the doctor

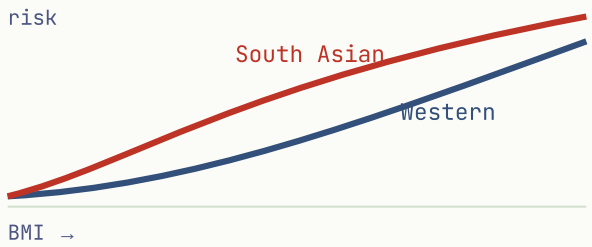
3 meals × 90 days

Post-meal spikes are where the damage happens, and they are invisible to a quarterly lab test. Care stays reactive.

Same diagnosis, different body, different plate.

PHENOTYPE

Insulin resistance at lower BMI



Our cohort: mean BMI 26.6, yet mean fasting glucose 150 mg/dL

PLATE

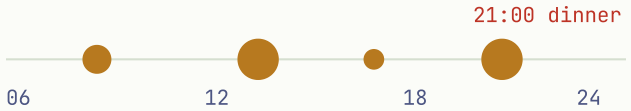
Rice, roti, dosa, chai, sweets

Masala dosa	68 g · GI 77
Rajma chawal	82 g · GI 68
Chicken biryani	90 g · GI 72
Ragi mudde + saaru	60 g · GI 55
Gulab jamun (2)	40 g · GI 80

38 regional dishes modelled with carbs and glycaemic index

ROUTINE

Late dinners, short sleep, festivals



Metformin missed on some days; one bad night lowers next-day insulin sensitivity

OUR ANSWER

One virtual patient, built from two data streams.

Static · EHR

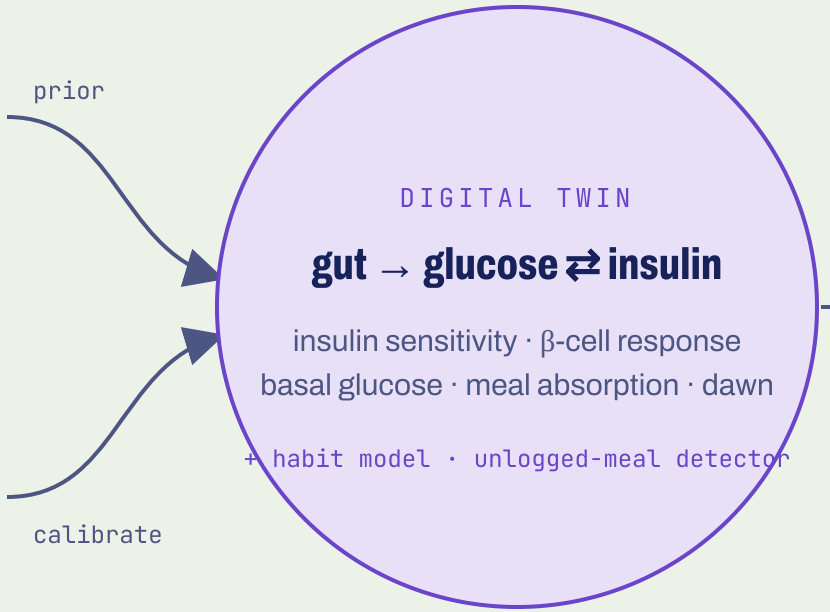
HbA1c, labs, diagnoses, drugs, BMI, TCF7L2 genotype

FHIR R4 · LOINC · SNOMED

Dynamic · wearables

CGM, steps, heart rate, HRV, sleep stages, meals, doses

every 5 minutes



Predict

spikes > 180 and lows < 70, 2 h ahead

Explain

meal, routine, sleep, activity, dose

Simulate

what if, then send it to the patient

DATA · SANDBOX RULES

A synthetic Indian cohort, built from physiology and made realistically messy.

150

patients across 13 cities
114 T2D · 36 prediabetes

604,800

five-minute wearable readings
over 14 days

0

real patients
DPDP Act and HIPAA safe by design

`CGM lag, noise, dropouts` `45–98% of meals logged` `carbs under-estimated` `missed metformin` `short sleep lowers insulin sensitivity` `festival day` `FHIR R4 export`

CGM is simulated from each patient's hidden physiology; HbA1c and fasting glucose are derived back from it, so both streams agree. The hidden truth is kept aside to check the twin.

CHAYA छाया · system architecture

Two data streams in, one calibrated virtual patient; 2-hour spike and low-glucose warnings, what-if simulation and a patient app out. Patient logs flow back in.

1 · DATA STREAMS

Static / historical: EHR

FHIR R4 Bundle per patient

- Demographics, city, diet
- Diagnoses (SNOMED): T2D, HTN, NAFLD
- Labs (LOINC): HbA1c, FPG, lipids, eGFR
- Medications: metformin, glimepiride, DPP-4i, SGLT2i, insulin
- Genetics: TCF7L2 rs7903146

Dynamic / real-time: wearables

- CGM glucose, every 5 min
- Steps, heart rate
- Nightly HRV (RMSSD), sleep stages
- Meal and dose logs, now from the patient app

Synthetic cohort

150 Indian patients × 14 days · 604,800 sensor rows · physiology-driven generator. No real patient data.

2 · INGEST & HARMONISE

FHIR parser

LOINC / SNOMED codes → patient profile vector

Stream alignment

5-min grid · sensor gaps kept as gaps · meal and dose events placed on the timeline

Model inputs

Minute-level carbs, GI-dependent absorption, steps, sleep-debt and missed-dose modifiers

3 · DIGITAL TWIN CORE

Per-patient physiological model

`gut → plasma glucose ⇄ insulin action`
`extended Bergman minimal model`

1. **EHR prior.** BMI, waist, HbA1c, duration, TCF7L2, drugs → starting physiology
2. **Calibrate.** MAP fit to 7 days of CGM: insulin sensitivity, β-cell response, basal glucose, meal bioavailability, dawn effect
3. **Observe.** Runs beside live CGM; corrects its state and detects unlogged meals
4. **Habit model.** Learns when and how much this patient usually eats
5. **Forecast & what-if.** Simulates the next 2–4 h under routine or a doctor's scenario

↻ re-calibrates as new data arrives

4 · FUSION & PREDICTION

Fused feature vector

every 15 min

`18 EHR` `25 wearable` `11 twin`

Spike classifier

LightGBM · P(glucose > 180 mg/dL within 2 h)

Glucose regressor

LightGBM · glucose value at +2 h

Low-glucose risk NEW

Twin forecast < 80 mg/dL in 2 h → *Low likely*; CGM < 70 → *Low now*. 7 of 10 lows warned, median 80 min ahead

Explanations

Tree SHAP → clinical drivers: meal, routine, sleep, activity, medication, physiology

5 · DELIVERY

Simulated clinic

- Pinned register, case sheet, stamps
- Reasons, twin physiology
- Escalation: owner, WhatsApp, call log
- Rx pad dock: what-if → save & send
- Alert threshold · Outcomes page

Patient app NEW

- English · हिंदी
- Glucose + 2 h shadow, plain warnings
- Dose reminders, meal logging
- Doctor's advice with read receipt

↺ meals and doses logged in the app feed the twin's inputs

Evaluation protocol

Patient-level split 105 train / 45 held-out. Temporal split: days 1–7 calibrate, days 8–14 predict. Threshold chosen on training folds only.

Privacy by design

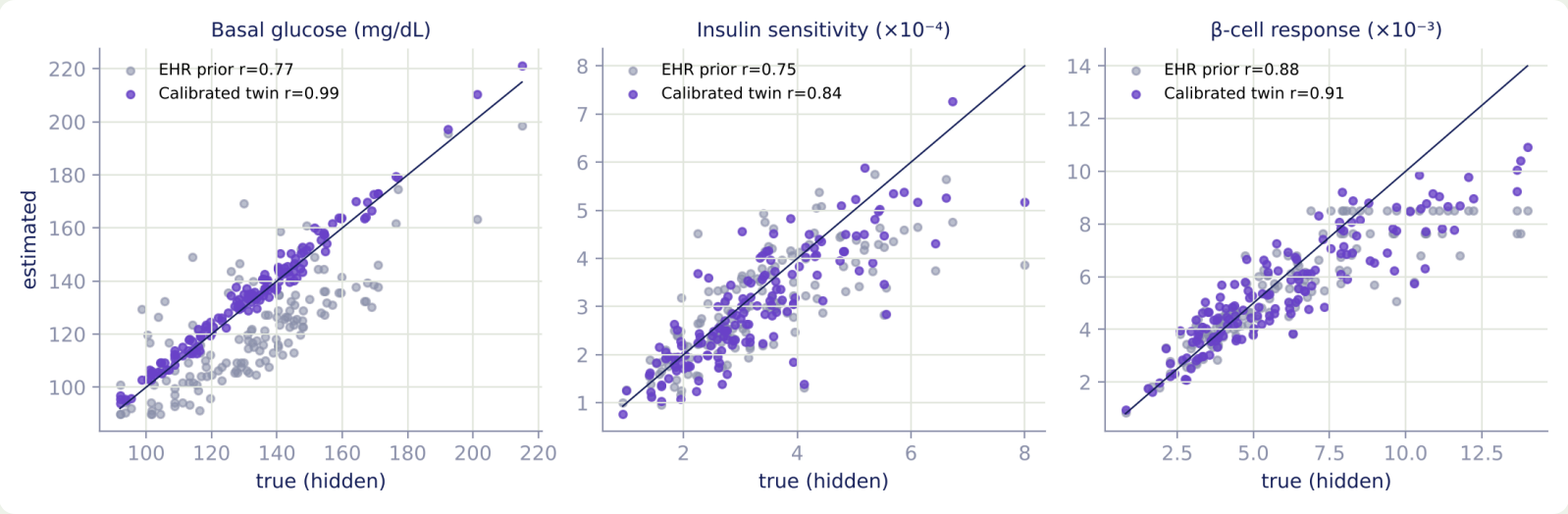
Synthetic data only (DPDP Act 2023 / HIPAA sandbox rules). FHIR-native so it can plug into ABDM-consented records later.

Stack

Python · NumPy/Numba · SciPy · LightGBM · FastAPI (local pipeline) · Vercel functions (Mumbai) · Supabase Postgres · vanilla JS + D3

DOES THE TWIN LEARN THE PERSON?

A week of wearable data recovers each patient's hidden physiology.



BASAL GLUCOSE

0.99 vs 0.77 EHR

INSULIN SENSITIVITY

0.84 vs 0.75 EHR

B-CELL RESPONSE

0.91 vs 0.88 EHR

Correlation with true hidden parameters · 150 patients

RESULTS · 45 HELD-OUT PATIENTS

Eight in ten spikes, flagged before they happen.

0.942
AUROC, spike > 180 mg/dL within 2 hours

83%
sensitivity at 79% precision

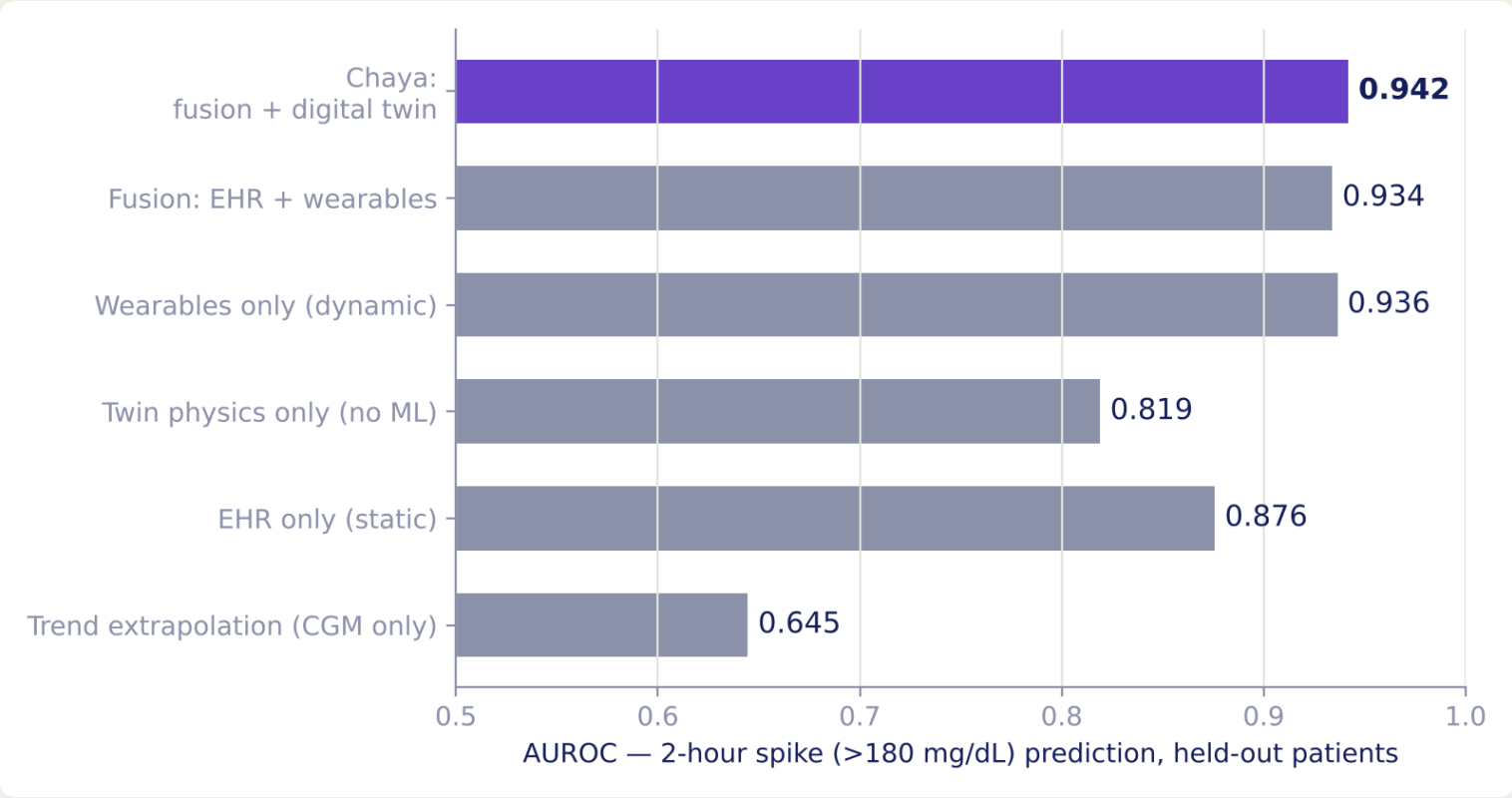
87%
of spikes flagged ≥ 60 min ahead

24
mg/dL error on 2-h glucose (53 if glucose is assumed flat)

Patient-level split: 105 train / 45 test. Days 1–7 calibrate, days 8–14 predict. Threshold set on training folds only. Clarke error grid A+B: 99.3%.

FUSION MATTERS

Each stream adds signal; the twin adds physiology.



Twin physics, zero training: 0.819, far above the CGM trend at 0.645. It works from day one.

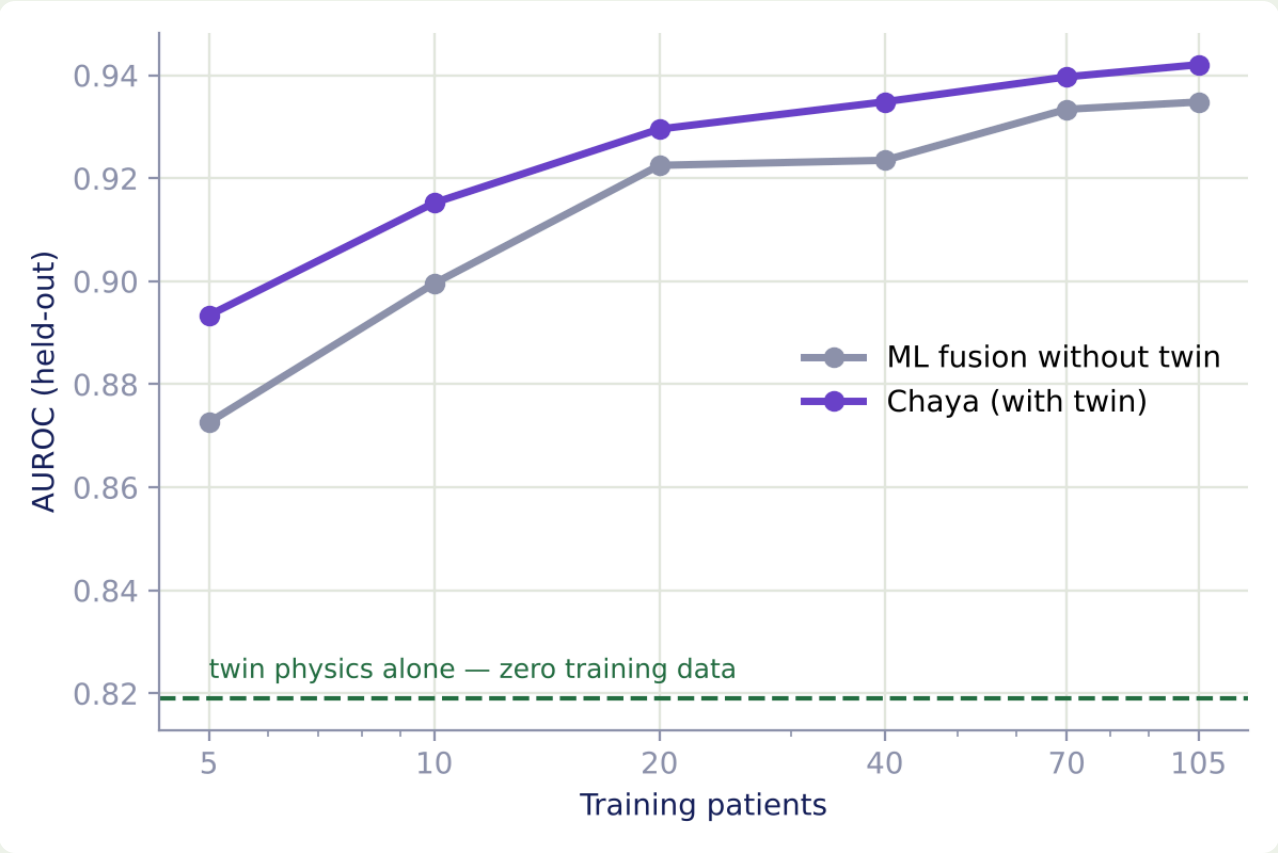
EHR alone knows who is at risk; **wearables** know when.

Fusion + twin gives the best of both: 0.942.

AUROC on held-out patients

WHEN THE TWIN MATTERS MOST

Physics fills the gap when data is scarce.



With only **5 training patients**, the twin lifts AUROC from `0.873` to `0.893`.

That is the situation of a **new clinic**, a **new patient** or a **rare situation** such as a festival day.

With lots of data, ML alone catches up. The twin's lasting value is what-if simulation, which a black box cannot do.

CHAYA छाया | Clinic | Patient app | Outcomes | Data sources | OFFLINE DEMO | Guide

SIMULATED DEMO · EVERY PATIENT IS SYNTHETIC · DECISION SUPPORT, NOT FOR CLINICAL USE

Mon 14 · Tue 15 · Wed 16 · Thu 17 · Fri 18 · Sat 19 · Sun 20 | NOW | 12:45 | Replay | Alerts: Balanced · 4.2/day

Replaying synthetic week · Mon 14 to Sun 20 Sep 2026

Register 45 of 45

Search name, city or ID

Most spike alerts this week

- **Pooja Rathore** — 58 warnings — 43F · Jaipur · T2D — A1c 8.0%
- **Suresh Jadhav** — 56 warnings — 41M · Pune · T2D — A1c 8.1%
- **Srinivas Goud** — 55 warnings — 60M · Hyderabad · T2D — A1c 8.7%
- **Fatima Begum** — 55 warnings — 52F · Bhubaneswar · T2D — A1c 8.3%
- **Deepak Menon** — 47 warnings — 42M · Kochi · T2D — A1c 7.9%
- **Rajesh Bose** — 46 warnings

GT-1102 · synthetic

Sanjay Krishnan

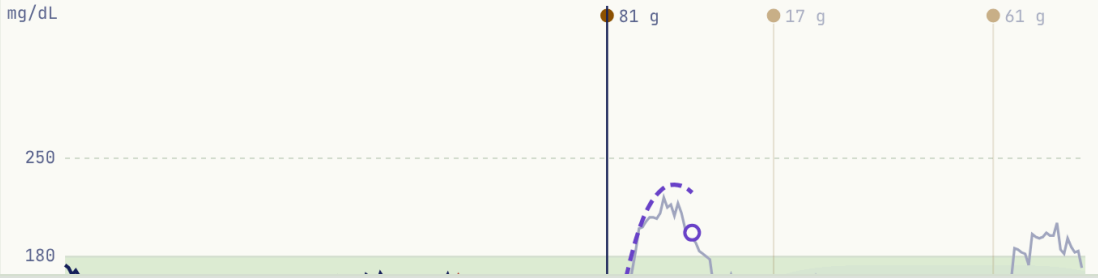
53 · Male · Chennai · Non-vegetarian · BMI 23.9

Type 2 diabetes · 3.9 y | Fatty liver | Metformin | Glimepiride

Sanjay's app | Data sources

TIME IN RANGE	ABOVE 180	GMI	VARIABILITY	SPIKE WARNINGS
86 %	14 %	6.9 %	18 % CV	26
meets goal of 70%	above 250: 0%	lab HbA1c 6.4%	stable (goal ≤ 36%)	this week, 2 h ahead

Day trace | Sources | Sunday 20 Sep



Next 2 hours | Sources | Sun 12:45

SPIKE LIKELY · 92% · NEXT 2 H

92% chance glucose goes above 180 mg/dL by 14:45.

GLUCOSE IN 2 H	FORECAST PEAK
197 mg/dL	231 at 14:25

WHY CHAYA SAYS THIS

RECENT MEAL
81 g carbs logged 0 min ago (GI 75)

TWIN FORECAST
Twin projects a peak of 231 mg/dL

ACTIVITY
251 steps in the last hour

Mark warning as reviewed

Who acts on this?

EXPLAINABLE BY DESIGN

Every alert comes with its reasons.

Next 2 hours (i) Sources Sun 12:45

SPIKE LIKELY
92% · NEXT 2 H

92% chance glucose goes above 180 mg/dL by **14:45**.

GLUCOSE IN 2 H	FORECAST PEAK
197 mg/dL	231 at 14:25

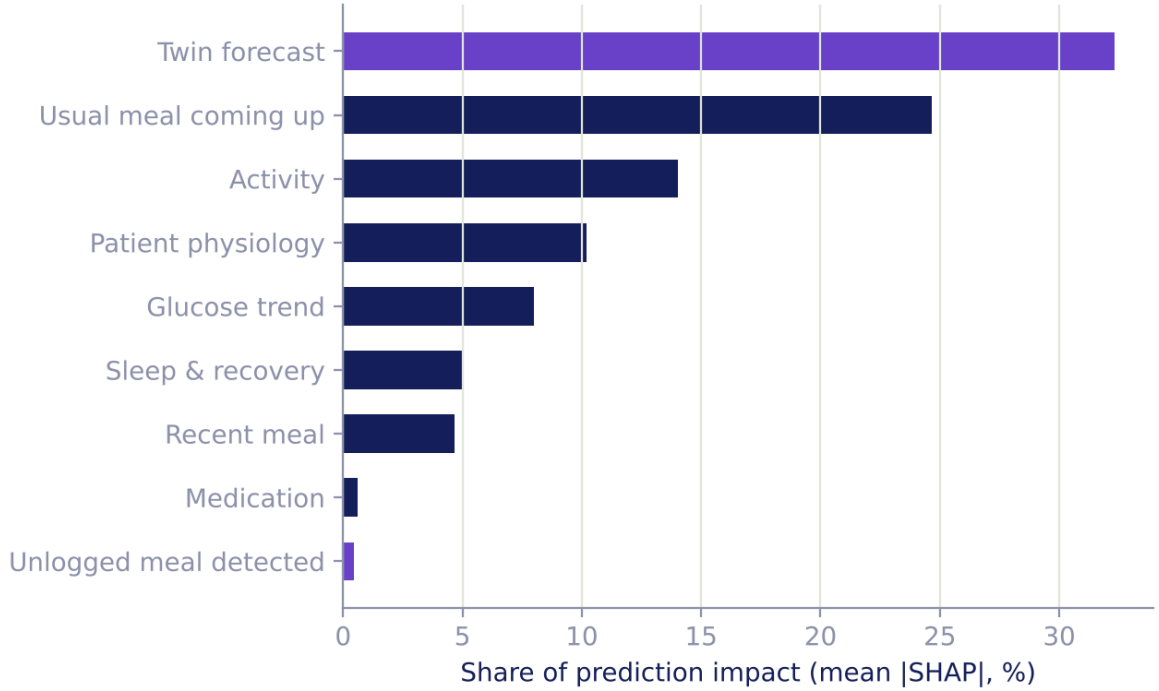
WHY CHAYA SAYS THIS (i)

RECENT MEAL
81 g carbs logged 0 min ago (GI 75)

TWIN FORECAST
Twin projects a peak of 231 mg/dL

ACTIVITY
251 steps in the last hour

Mark warning as reviewed



Tree SHAP contributions grouped into the drivers a clinician thinks in. Violet: drivers that come from the twin.

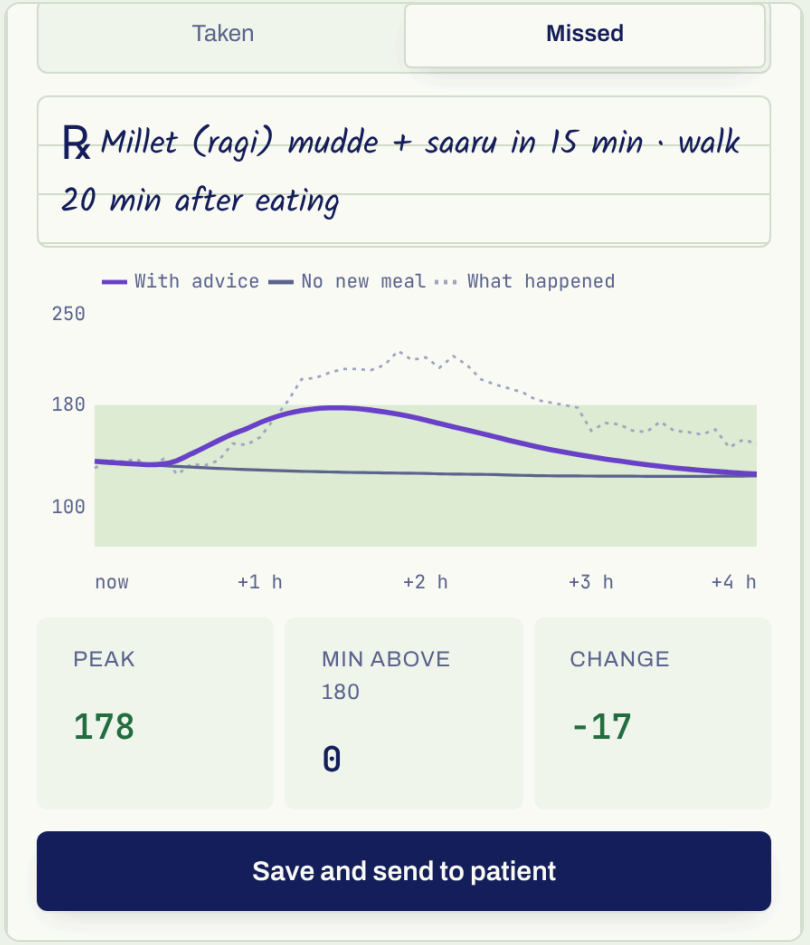
WHAT-IF SIMULATION

Advice in numbers, for this patient's body.

SANJAY · 53 · T2D 3.9 Y · SUNDAY 12:15, BEFORE LUNCH

LUNCH IN 15 MIN	PEAK MG/DL	TIME > 180
Chicken biryani 90 g · GI 72	237	115 min
Biryani + 20-min walk	217	85 min
Ragi mudde + saaru 60 g · GI 55	194	70 min
Ragi mudde + 20-min walk	178	0 min

Same patient, same morning. Swapping the grain and walking after lunch keeps him in range.



Peak glucose over the next 4 h, simulated by the calibrated twin

SAFETY · NEW

The same twin, read downwards, warns of lows.

7 of 10

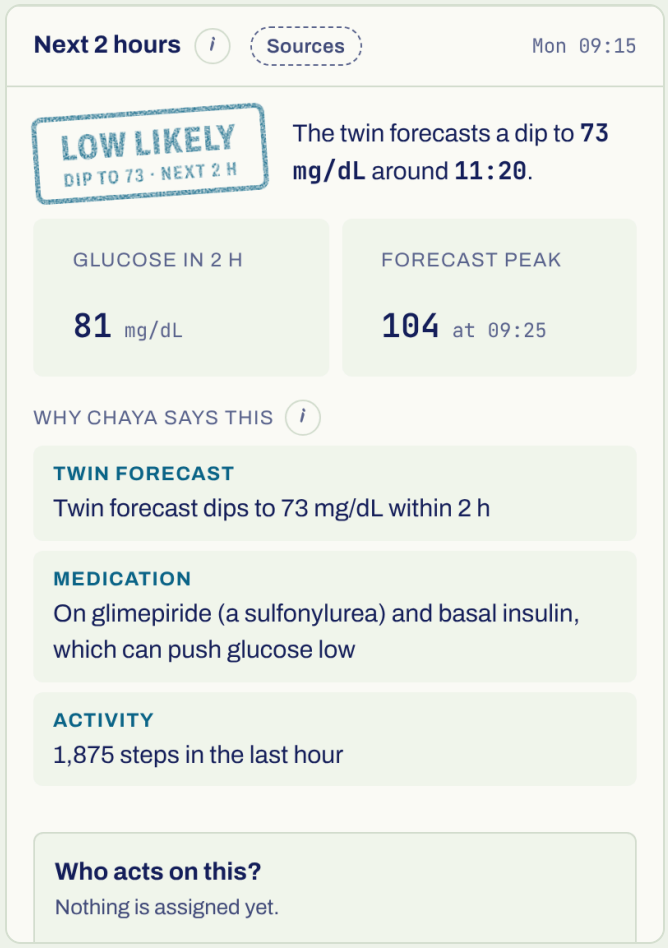
held-out lows warned within 2 h

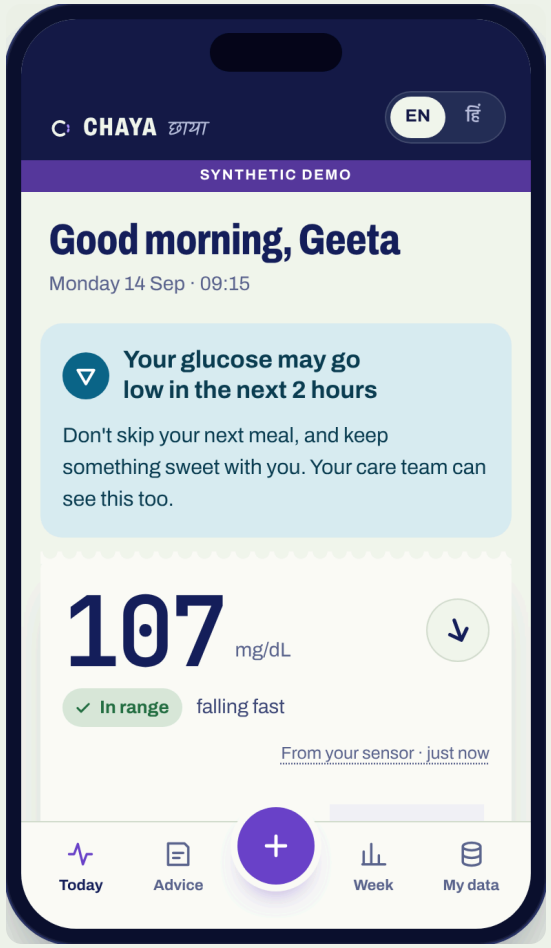
80 min

median warning before glucose fell below 70

0.16

low warnings per patient a day, about one a week

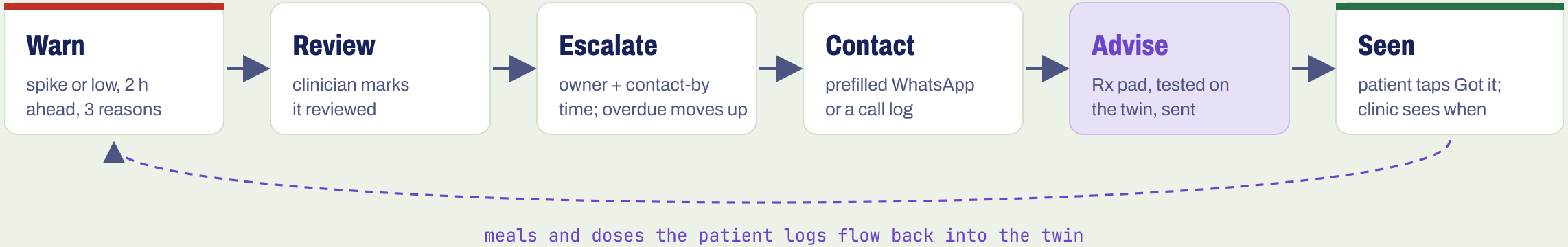




Forecast below 80 mg/dL → Low likely · CGM below 70 → Low now · Geeta Mukherjee, on glimepiride and basal insulin

FROM WARNING TO ACTION · NEW

Every warning gets an owner, a deadline and a way to reach the patient.



Doctor on call

Due Mon 09:30 · 15 min left

WhatsApp | Log a call

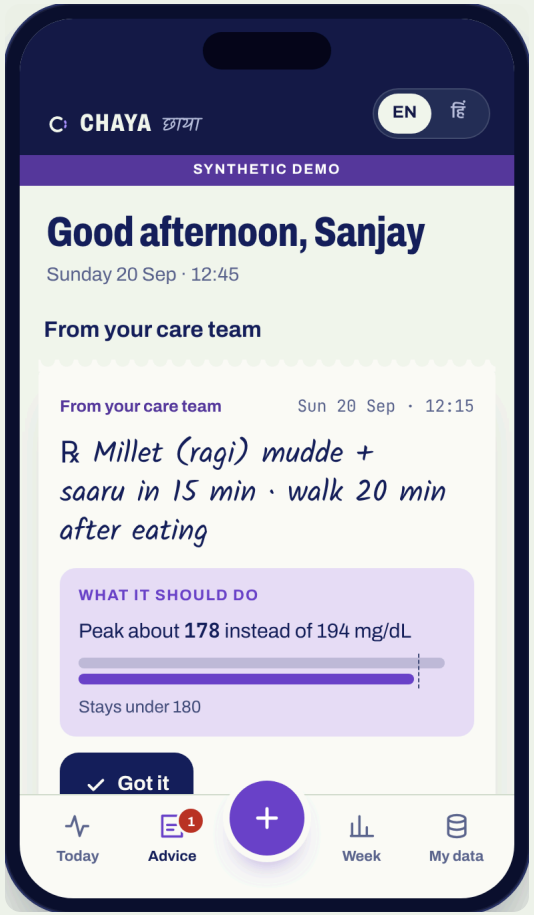
Owners: care coordinator, doctor on call or diabetes educator.

Stored in Supabase: `alert_reviews` · `escalations` · `case_notes` · `patient_logs`

The patient carries the carbon copy, in English or Hindi.



TODAY · GLUCOSE + 2 H SHADOW



ADVICE · 178, NOT 194



हिंदी · ONE TAP

Dose reminders at the patient's own usual times.

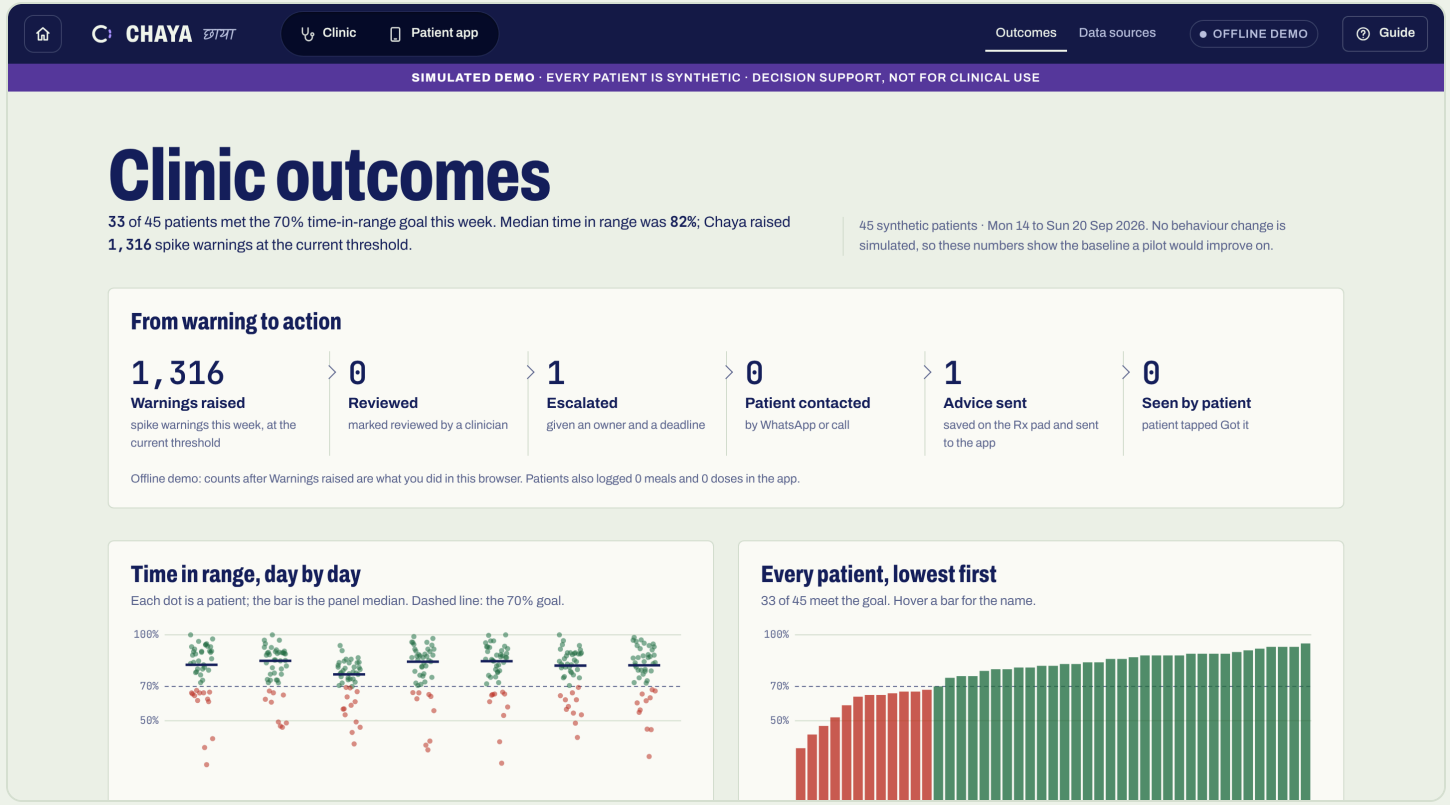
Log a meal from 38 Indian dishes or a photo; it lands on the clinic trace.

Lows first: sugar rule, recheck, call.

No risk scores, no twin parameters, no other patients.

FOR THE HOSPITAL BUYER · NEW

Clinic outcomes, and an alert threshold the clinic controls.

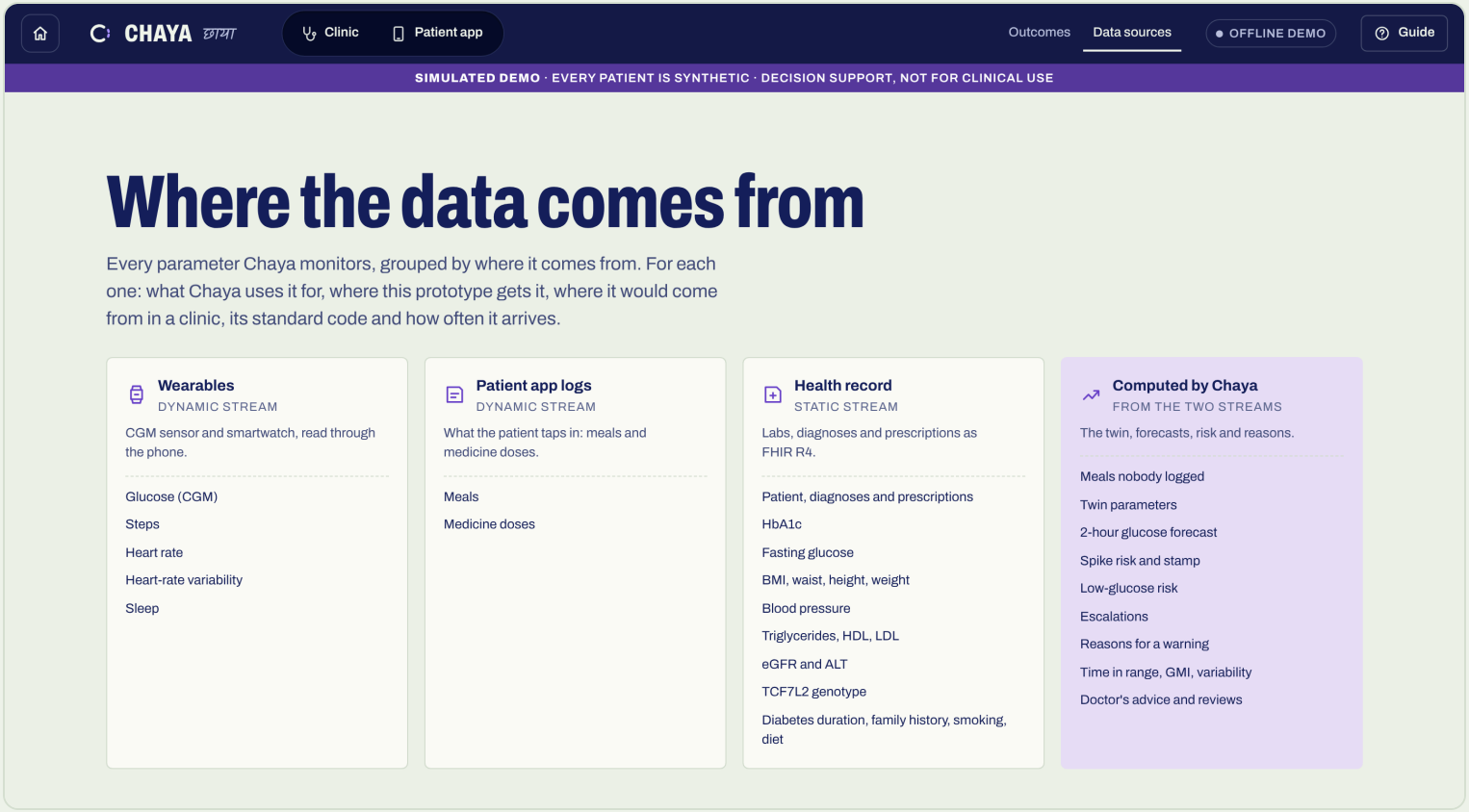




2.6 → 4.6 warnings per patient a day buys **74% → 98%** of spikes warned within 2 h.

TRANSPARENCY · NEW

Every number has a source you can name.



25

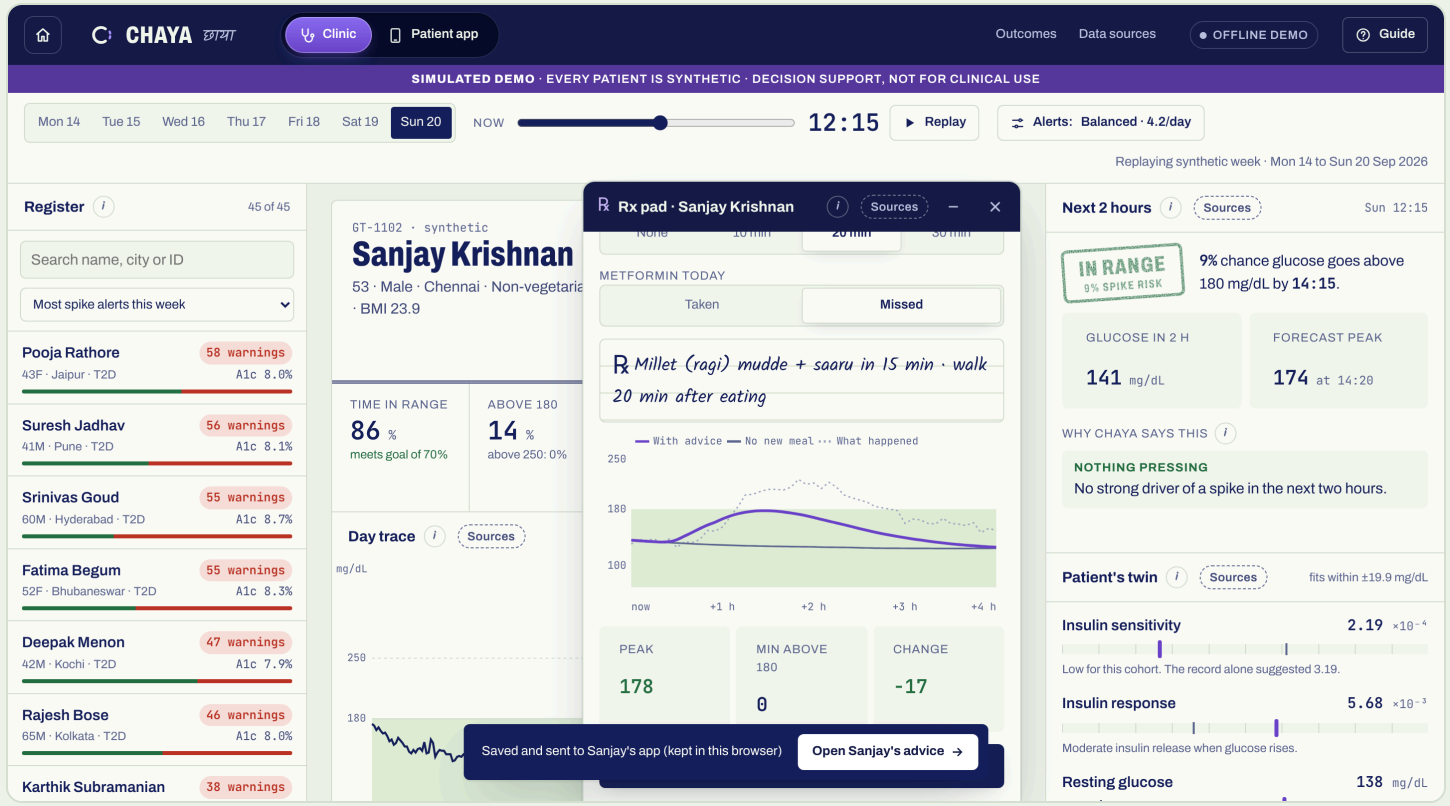
parameters in four groups: wearables, patient app logs, health record, computed by Chaya

For each: what it is used for, where the prototype gets it, where a clinic would get it, its **LOINC / SNOMED / FHIR** code and cadence.

A Sources button on every panel opens the same detail.

BUILT FOR THE CLINIC'S DAY · NEW

Advice is one tap away, and lands where the patient will see it.



Pinned panes: register on the left edge, forecast and twin on the right.

℞ Try advice opens the Rx pad like an email draft; minimise or close.

Save and send shows a toast that opens that patient's Advice tab.

Clear navigation: home → clinic or patient app; inside, a switcher that keeps the same patient.

Three guided tours: product, Sanjay's case, patient app.

Unique names: every patient has a surname.

SAFETY AND PRIVACY

Built to support the doctor, not to replace them.

Clinician in the loop

Decision support only. No dosing. Every alert shows its reasons and the twin's projection.

Alert fatigue managed

Consecutive alerts merge into one episode. The clinic sets the threshold and sees the trade-off: 2.6–4.6 per patient-day.

Lows come first

Low warnings lead the patient's screen with the 15 g sugar rule; sulfonylurea and insulin users are flagged.

DPDP-ready

Synthetic data only. FHIR-native, so it can use ABDM-consented records with purpose limitation.

NEXT

From prototype to clinic.

Now

Clinic + bilingual patient app, spikes and lows, live on Vercel + Supabase

Validate

Public CGM + wearable datasets (CGMacros, BIG IDEAs Lab)

Pilot

Supervised pilot with a partner diabetes clinic

Extend

WhatsApp Business API, ABDM-consented records, hypertension twin

PROACTIVE, NOT REACTIVE

See the spike before it happens. Make sure someone acts. Show the patient how to avoid it.

chaya-tau.vercel.app · github.com/r-e-sai-harsha/TheGeeks_MastersUnion

The Geeks · R. E. Sai Harsha Sadhu · Sanskriti Saraf · Muskan Sharma